

القيم - جيب - جيب التمام

$$\int \tan x dx = \ln |\sec x|$$

$$\int \cot x dx = \ln |\sin x|$$

$$\int \csc x dx = -\ln |\csc x + \cot x|$$

$$\int \sec x dx = \ln |\sec x + \tan x|$$

$$5) \cos^2 x = \frac{1}{2}(1 + \cos(2x)) \quad u = \sin x$$

$$6) \sin^2 x = \frac{1}{2}(1 - \cos(2x)) \quad u = \cos x$$

$$7) \sin(2x) = 2 \sin x \cos x$$

$$\int \sin(ax) dx = -\frac{1}{a} \cos(ax) + C$$

$$\int \cos(ax) dx = \frac{1}{a} \sin(ax) + C$$

$$10) \sec^2 x = 1 + \tan^2 x \quad u = \tan x$$

$$11) \tan^2 x = \sec^2 x - 1 \quad u = \sec x$$

$$12) \csc^2 x = 1 + \cot^2 x \quad u = \cot$$

$$13) \cot^2 x = \csc^2 x - 1 \quad u = \csc x$$

$$14) \sin^2 x + \cos^2 x = 1$$

$$15) \sqrt{9-x^2} = x = 3 \sin \theta$$

$$16) \sqrt{2^2-x^2} = x = 2 \tan \theta, 2 \sin \theta$$

$$17) \sqrt{x^2-1} = x = \sec \theta$$

$$18) (4x^2+9)^{3/2} = 2x = 3 \tan \theta$$

$$19) \int \frac{dx}{\sqrt{a^2-x^2}} = \sin^{-1}\left(\frac{x}{a}\right) + C$$

$$20) \int \frac{x}{\sqrt{a^2-x^2}} dx = -\sqrt{a^2-x^2} + C$$

$$21) \int \frac{dx}{\sqrt{x^2 \pm a^2}} = \ln|x + \sqrt{x^2 \pm a^2}| + C$$

$$22) \int \frac{x}{\sqrt{x^2 \pm a^2}} dx = \sqrt{x^2 \pm a^2} + C$$

$$23) \int \frac{dx}{x^2+a^2} = \frac{1}{a} \tan^{-1}\left(\frac{x}{a}\right) + C$$

$$24) \sin x = \frac{2 \tan(x/2)}{1 + \tan^2(x/2)}$$

$$25) \cos x = \frac{1 - \tan^2(x/2)}{1 + \tan^2(x/2)}$$

$$26) \sin x = \frac{2u}{1+u^2}$$

$$27) \cos x = \frac{1-u^2}{1+u^2}$$

$$28) dx = \frac{2}{1+u^2} du$$

$$29) \int x^n dx = \frac{x^{n+1}}{n+1} + C$$

$$30) \int \frac{1}{x} dx = \ln|x| + C$$

$$31) \int e^x dx = e^x + C$$

$$32) \int a^x dx = \frac{a^x}{\ln(a)} + C$$

$$33) \int \frac{dx}{a^2+x^2} = \frac{1}{a} \tan^{-1}\left(\frac{x}{a}\right) + C$$

$$34) \int \frac{dx}{x^2-a^2} = \frac{1}{2a} \ln \left| \frac{x-a}{x+a} \right| + C$$

$$35) \int \frac{x}{x^2 \pm a^2} = \frac{1}{2} \ln |x^2 \pm a^2| + C$$

$$36) \int \frac{dx}{\sqrt{a^2-x^2}} = \sin^{-1}\left(\frac{x}{a}\right) + C$$

$$37) \int \frac{dx}{\sqrt{x^2 \pm a^2}} = \ln |x + \sqrt{x^2 \pm a^2}| + C$$

$$38) \int \frac{x}{\sqrt{x^2 \pm a^2}} dx = \sqrt{x^2 \pm a^2} + C$$

$$39) \int \frac{x}{\sqrt{a^2-x^2}} dx = -\sqrt{a^2-x^2} + C$$

$$40) \int \frac{1}{x\sqrt{x^2-a^2}} dx = \frac{1}{a} \sec^{-1}\left(\frac{x}{a}\right) + C$$

For Limit #4.4

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Log & Ln		Exponential function	
$\ln 0^+$	$-\infty$	e^0	1
$\log_a 0^+$		a^0	
$\ln \infty$	∞	$e^{-\infty}$	0
$\log_a \infty$		$a^{-\infty}$	
$\ln 1$	0	e^{∞}	∞
$\log_a 1$		a^{∞}	
$\ln 0$	1	$\frac{1}{\infty} = 0$	$\frac{1}{0} = \infty$
$\log_a a$ <i>نفسه</i>			
Inverse Trig. Function		Trig. Function	
$\tan^{-1}(0)$	0	$\tan(0)$	0
$\sin^{-1}(0)$		$\sin(0)$	
$\cos^{-1}(1)$			
$\sin^{-1}(1)$	$\frac{\pi}{2}$ <i>90°</i>	$\tan(\pi/4)$	1
$\cos^{-1}(0)$		$\cos(0)$ <i>sec(0)</i>	
$\tan^{-1}(\pm\infty)$	$\pm\frac{\pi}{2}$	$\sin(\pi/2)$	∞
<i>$\cos^{-1}, \sin^{-1} = \text{undefined}$</i>		$\csc(0)$	
		$\sec(\pi/2)$	
$\tan^{-1}(1)$	$\frac{\pi}{4}$	$\cos(\pi/2)$	0
Hyperbolic Function			
$\tanh(0)$	0	$\sinh(\infty)$	∞
$\sinh(0)$		$\cosh(\infty)$	
$\cosh(0)$	1	$\tanh(\infty)$	1

$$\textcircled{1} \cos\left(\frac{2\pi}{3}\right) = \frac{1}{2}$$

$$\textcircled{2} \sin\left(\frac{2\pi}{3}\right) = \frac{\sqrt{3}}{2}$$

$$\textcircled{3} \cos\left(\frac{\pi}{4}\right) = \frac{\sqrt{2}}{2}$$

$$\textcircled{4} \sin\left(\frac{\pi}{4}\right) = \frac{\sqrt{2}}{2}$$

$$\textcircled{5} \tan\left(\frac{\pi}{4}\right) = 1$$

$$\textcircled{6} \sin\left(\frac{\pi}{2}\right) = 1$$

$$\textcircled{7} \cos\left(\frac{\pi}{2}\right) = 0$$

$$\textcircled{8} \sin(\pi, 0, 2\pi, 3\pi, \dots) = 0$$

$$\textcircled{9} \cos\left(\frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \dots\right) = 0$$

$$\textcircled{10} \sin\left(\frac{\pi}{6}\right) = \frac{1}{2}$$

$$\textcircled{11} \cos\left(\frac{\pi}{3}\right) = \frac{1}{2}$$